In The Name of God



Sharif University of Technology
Electrical Engineering Department

Electronics 3

Simulation Project 2

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Problem 1

Figure 1 depicts a two-stage differential amplifier.

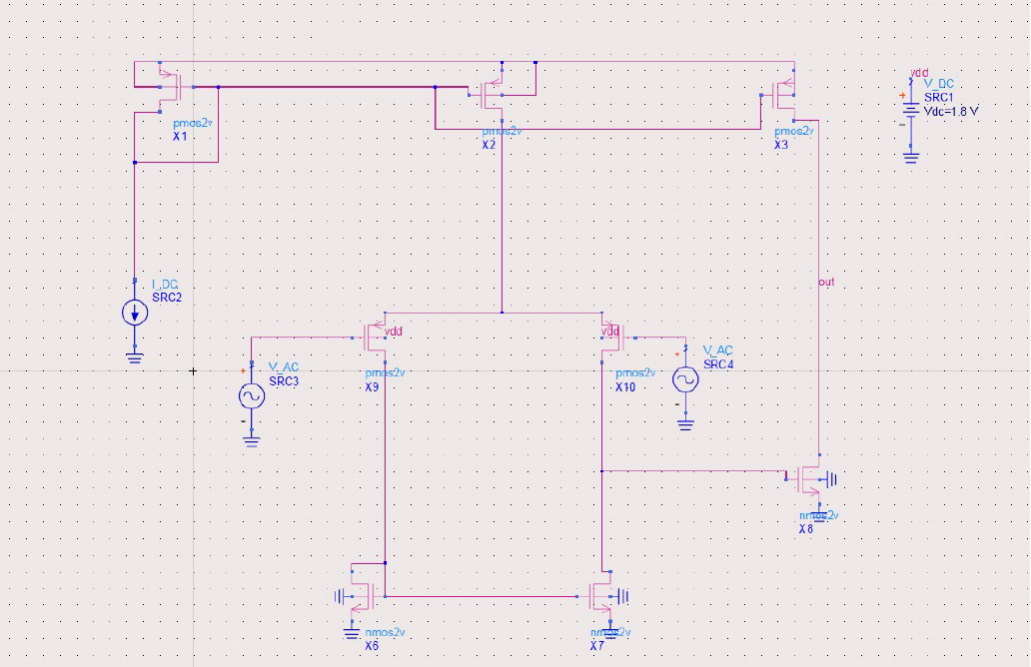


Figure 1: Schematic of the two-stage differential amplifier.

a. Design this circuit and calculate all necessary parameters to meet the following specifications:

- Voltage gain: $A_v > 60\,\text{dB}$
- Power consumption: $< 1\,\text{mW}$
- Common-mode range (CMR): $> 0.5\,\text{V}$

The first step is to calculate the voltage gain.

$$A_v = A_{v_1} \times A_{v_2} = g_{m_4}(r_{o_5}||r_{o_6}) \times g_{m_8}(r_{o_3}||r_{o_8})$$

According to the Power consumption limit the maximum total current flowing in the circuit is:

$$I_{tot_{max}} = \frac{P_{max}}{V_{DD}} = 0.555^{mA}$$

So the sum of I_1, I_2 and I_3 shouldn't exceed 0.555^{mA}.

g_{m_4} and g_{m_8} need to be large so I_4 and I_3 shouldn't be small. I set $I_3 = 0.255^{mA}$ and also $I_2 = 2I_4 = 2I_5 = 0.255^{mA}$. So because of the total current limit $I_1 \leq 0.045^{mA}$. I set $I_1 = 0.04^{mA}$ and according to the currents determined the W and L parameters of each transistor.

We should note that $g_m = \sqrt{2I_D\mu C_{ox}\frac{W}{L}}$ so to have a greater g_m for a transistor we need to increase $\frac{W}{L}$ of the device.

At first the gain requirement was not met because some of the transistors were off. This

problem was the solved by adjusting the amount of $\frac{W}{L}$ of those transistors. I found out that increasing the $\frac{W}{L}$ of M_8 and also M_4 and M_5 helps to improve the gain. But at some point if I kept increasing the $\frac{W}{L}$ of those transistors the gain would decrease remarkably since they were off or in triode. After some adjustments in the $\frac{W}{L}$s I finally found the right numbers. The designed circuit is shown in figure 2.

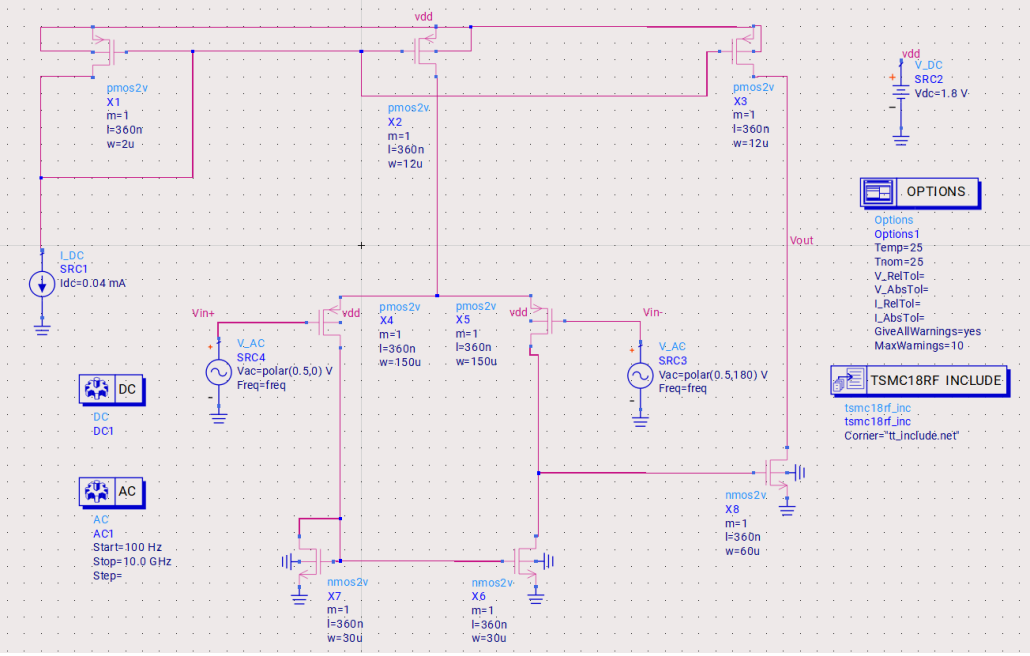


Figure 2: The designed circuit for part a

The gain and power conditions are investigated in the following figures.

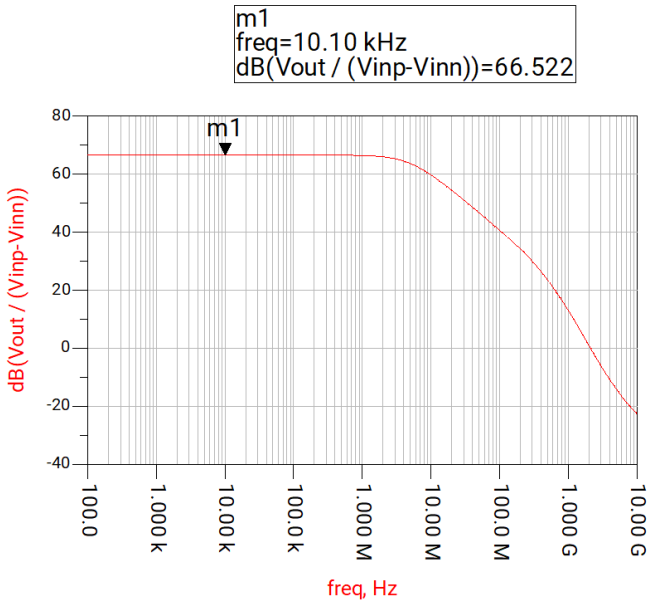


Figure 3: The gain of the designed circuit

As you can see in 4 all the transistors are active and in saturation region. Also the circuit consumes a total power of 0.998^{mW} which is acceptable.

index	V_Source.DC_DCOP.Power	index	Id	Vds-(Vgs-Vth)	Vgs-Vth
1	-998.1 u	1	-40.00 u	-471.7 m	-482.0 m
2	0.0000	2	-255.4 u	-512.5 m	-479.8 m
3	0.0000	3	-259.1 u	-704.8 m	-480.0 m
		4	-127.7 u	-209.2 m	-63.51 m
		5	-127.7 u	-209.2 m	-63.51 m
		6	127.7 u	493.0 m	41.97 m
		7	127.7 u	493.0 m	41.97 m
		8	259.1 u	573.1 m	42.04 m

Figure 4: The operating points of the transistors and the power consumption

b. Calculate the CMR of your circuit using *ADS* and verify whether your circuit meets the specified conditions or not.

In order to check the Common-mode input range of the circuit I swept the input voltage from -0.5^V to 1.8^V. To do this I used *Variable Equation* and *Parameter Sweep* boxes.

Then I plotted the gain of the circuit vs. the input common-mode voltage.

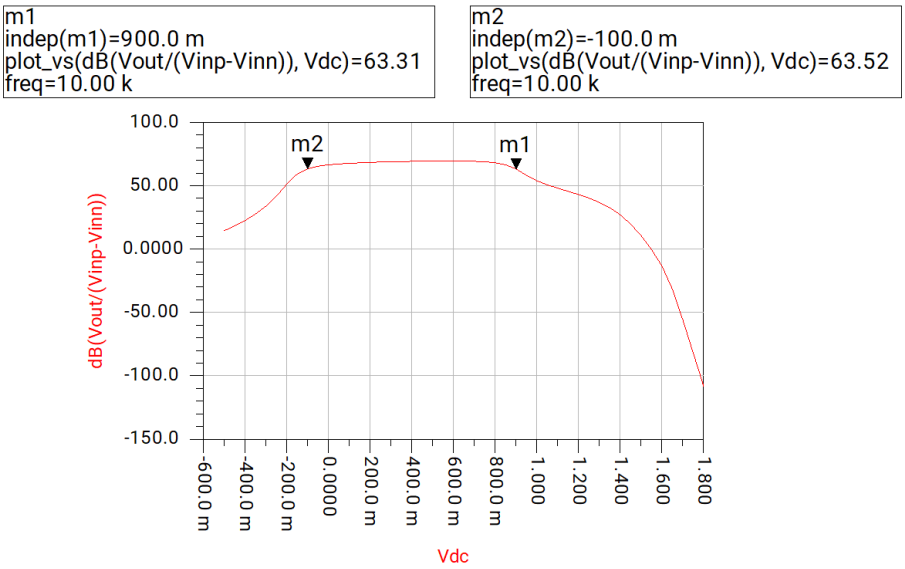


Figure 5: CMR analysis

As it can be observed in the preceding figure, the gain stays above $60dB$ for an input voltage range of $[-0.1^V\ ,\ 0.9^V]$. So the CMR of the circuit is nearly 1^V which meets our need.

In the next page you can check that the transistors all stay in saturation for the mentioned common-mode range.

Vdc	Id	Vds-(Vgs-Vth)	Vgs-Vth
index=1			
-500.0 m	-40.00 u	-471.7 m	-482.0 m
-400.0 m	-40.00 u	-471.7 m	-482.0 m
-300.0 m	-40.00 u	-471.7 m	-482.0 m
-200.0 m	-40.00 u	-471.7 m	-482.0 m
-100.0 m	-40.00 u	-471.7 m	-482.0 m
-27.76 a	-40.00 u	-471.7 m	-482.0 m
100.0 m	-40.00 u	-471.7 m	-482.0 m
200.0 m	-40.00 u	-471.7 m	-482.0 m
300.0 m	-40.00 u	-471.7 m	-482.0 m
400.0 m	-40.00 u	-471.7 m	-482.0 m
500.0 m	-40.00 u	-471.7 m	-482.0 m
600.0 m	-40.00 u	-471.7 m	-482.0 m
700.0 m	-40.00 u	-471.7 m	-482.0 m
800.0 m	-40.00 u	-471.7 m	-482.0 m
900.0 m	-40.00 u	-471.7 m	-482.0 m
1000.0 m	-40.00 u	-471.7 m	-482.0 m
1.100	-40.00 u	-471.7 m	-482.0 m
1.200	-40.00 u	-471.7 m	-482.0 m
1.300	-40.00 u	-471.7 m	-482.0 m
1.400	-40.00 u	-471.7 m	-482.0 m
1.500	-40.00 u	-471.7 m	-482.0 m
1.600	-40.00 u	-471.7 m	-482.0 m
1.700	-40.00 u	-471.7 m	-482.0 m
1.800	-40.00 u	-471.7 m	-482.0 m

index=2				index=3			
-500.0 m	-255.4 u	-512.5 m	-479.8 m	-500.0 m	-259.1 u	-704.8 m	-480.0 m
-400.0 m	-255.4 u	-512.5 m	-479.8 m	-400.0 m	-259.1 u	-704.8 m	-480.0 m
-300.0 m	-255.4 u	-512.5 m	-479.8 m	-300.0 m	-259.1 u	-704.8 m	-480.0 m
-200.0 m	-255.4 u	-512.5 m	-479.8 m	-200.0 m	-259.1 u	-704.8 m	-480.0 m
-100.0 m	-255.4 u	-512.5 m	-479.8 m	-100.0 m	-259.1 u	-704.8 m	-480.0 m
-27.76 a	-255.4 u	-512.5 m	-479.8 m	-27.76 a	-259.1 u	-704.8 m	-480.0 m
100.0 m	-255.4 u	-512.5 m	-479.8 m	100.0 m	-259.1 u	-704.8 m	-480.0 m
200.0 m	-255.4 u	-512.5 m	-479.8 m	200.0 m	-259.1 u	-704.8 m	-480.0 m
300.0 m	-255.4 u	-512.5 m	-479.8 m	300.0 m	-259.1 u	-704.8 m	-480.0 m
400.0 m	-255.4 u	-512.5 m	-479.8 m	400.0 m	-259.1 u	-704.8 m	-480.0 m
500.0 m	-255.4 u	-512.5 m	-479.8 m	500.0 m	-259.1 u	-704.8 m	-480.0 m
600.0 m	-255.4 u	-512.5 m	-479.8 m	600.0 m	-259.1 u	-704.8 m	-480.0 m
700.0 m	-255.4 u	-512.5 m	-479.8 m	700.0 m	-259.1 u	-704.8 m	-480.0 m
800.0 m	-255.4 u	-512.5 m	-479.8 m	800.0 m	-259.1 u	-704.8 m	-480.0 m
900.0 m	-255.4 u	-512.5 m	-479.8 m	900.0 m	-259.1 u	-704.8 m	-480.0 m
1000.0 m	-255.4 u	-512.5 m	-479.8 m	1000.0 m	-259.1 u	-704.8 m	-480.0 m
1.100	-255.4 u	-512.5 m	-479.8 m	1.100	-259.1 u	-704.8 m	-480.0 m
1.200	-255.4 u	-512.5 m	-479.8 m	1.200	-259.1 u	-704.8 m	-480.0 m
1.300	-255.4 u	-512.5 m	-479.8 m	1.300	-259.1 u	-704.8 m	-480.0 m
1.400	-255.4 u	-512.5 m	-479.8 m	1.400	-259.1 u	-704.8 m	-480.0 m
1.500	-255.4 u	-512.5 m	-479.8 m	1.500	-259.1 u	-704.8 m	-480.0 m
1.600	-255.4 u	-512.5 m	-479.8 m	1.600	-259.1 u	-704.8 m	-480.0 m
1.700	-255.4 u	-512.5 m	-479.8 m	1.700	-259.1 u	-704.8 m	-480.0 m
1.800	-255.4 u	-512.5 m	-479.8 m	1.800	-259.1 u	-704.8 m	-480.0 m

index=4				index=5				index=6			
-500.0 m	-127.7 u	-209.2 m	-63.51 m	-500.0 m	-127.7 u	-209.2 m	-63.51 m	-500.0 m	127.7 u	493.0 m	41.97 m
-400.0 m	-127.7 u	-209.2 m	-63.51 m	-400.0 m	-127.7 u	-209.2 m	-63.51 m	-400.0 m	127.7 u	493.0 m	41.97 m
-300.0 m	-127.7 u	-209.2 m	-63.51 m	-300.0 m	-127.7 u	-209.2 m	-63.51 m	-300.0 m	127.7 u	493.0 m	41.97 m
-200.0 m	-127.7 u	-209.2 m	-63.51 m	-200.0 m	-127.7 u	-209.2 m	-63.51 m	-200.0 m	127.7 u	493.0 m	41.97 m
-100.0 m	-127.7 u	-209.2 m	-63.51 m	-100.0 m	-127.7 u	-209.2 m	-63.51 m	-100.0 m	127.7 u	493.0 m	41.97 m
-27.76 a	-127.7 u	-209.2 m	-63.51 m	-27.76 a	-127.7 u	-209.2 m	-63.51 m	-27.76 a	127.7 u	493.0 m	41.97 m
100.0 m	-127.7 u	-209.2 m	-63.51 m	100.0 m	-127.7 u	-209.2 m	-63.51 m	100.0 m	127.7 u	493.0 m	41.97 m
200.0 m	-127.7 u	-209.2 m	-63.51 m	200.0 m	-127.7 u	-209.2 m	-63.51 m	200.0 m	127.7 u	493.0 m	41.97 m
300.0 m	-127.7 u	-209.2 m	-63.51 m	300.0 m	-127.7 u	-209.2 m	-63.51 m	300.0 m	127.7 u	493.0 m	41.97 m
400.0 m	-127.7 u	-209.2 m	-63.51 m	400.0 m	-127.7 u	-209.2 m	-63.51 m	400.0 m	127.7 u	493.0 m	41.97 m
500.0 m	-127.7 u	-209.2 m	-63.51 m	500.0 m	-127.7 u	-209.2 m	-63.51 m	500.0 m	127.7 u	493.0 m	41.97 m
600.0 m	-127.7 u	-209.2 m	-63.51 m	600.0 m	-127.7 u	-209.2 m	-63.51 m	600.0 m	127.7 u	493.0 m	41.97 m
700.0 m	-127.7 u	-209.2 m	-63.51 m	700.0 m	-127.7 u	-209.2 m	-63.51 m	700.0 m	127.7 u	493.0 m	41.97 m
800.0 m	-127.7 u	-209.2 m	-63.51 m	800.0 m	-127.7 u	-209.2 m	-63.51 m	800.0 m	127.7 u	493.0 m	41.97 m
900.0 m	-127.7 u	-209.2 m	-63.51 m	900.0 m	-127.7 u	-209.2 m	-63.51 m	900.0 m	127.7 u	493.0 m	41.97 m
1000.0 m	-127.7 u	-209.2 m	-63.51 m	1000.0 m	-127.7 u	-209.2 m	-63.51 m	1000.0 m	127.7 u	493.0 m	41.97 m
1.100	-127.7 u	-209.2 m	-63.51 m	1.100	-127.7 u	-209.2 m	-63.51 m	1.100	127.7 u	493.0 m	41.97 m
1.200	-127.7 u	-209.2 m	-63.51 m	1.200	-127.7 u	-209.2 m	-63.51 m	1.200	127.7 u	493.0 m	41.97 m
1.300	-127.7 u	-209.2 m	-63.51 m	1.300	-127.7 u	-209.2 m	-63.51 m	1.300	127.7 u	493.0 m	41.97 m
1.400	-127.7 u	-209.2 m	-63.51 m	1.400	-127.7 u	-209.2 m	-63.51 m	1.400	127.7 u	493.0 m	41.97 m
1.500	-127.7 u	-209.2 m	-63.51 m	1.500	-127.7 u	-209.2 m	-63.51 m	1.500	127.7 u	493.0 m	41.97 m
1.600	-127.7 u	-209.2 m	-63.51 m	1.600	-127.7 u	-209.2 m	-63.51 m	1.600	127.7 u	493.0 m	41.97 m
1.700	-127.7 u	-209.2 m	-63.51 m	1.700	-127.7 u	-209.2 m	-63.51 m	1.700	127.7 u	493.0 m	41.97 m
1.800	-127.7 u	-209.2 m	-63.51 m	1.800	-127.7 u	-209.2 m	-63.51 m	1.800	127.7 u	493.0 m	41.97 m

index=7				index=8			
-500.0 m	127.7 u	493.0 m	41.97 m	-500.0 m	259.1 u	573.1 m	42.04 m
-400.0 m	127.7 u	493.0 m	41.97 m	-400.0 m	259.1 u	573.1 m	42.04 m
-300.0 m	127.7 u	493.0 m	41.97 m	-300.0 m	259.1 u	573.1 m	42.04 m
-200.0 m	127.7 u	493.0 m	41.97 m	-200.0 m	259.1 u	573.1 m	42.04 m
-100.0 m	127.7 u	493.0 m	41.97 m	-100.0 m	259.1 u	573.1 m	42.04 m
-27.76 a	127.7 u	493.0 m	41.97 m	-27.76 a	259.1 u	573.1 m	42.04 m
100.0 m	127.7 u	493.0 m	41.97 m	100.0 m	259.1 u	573.1 m	42.04 m
200.0 m	127.7 u	493.0 m	41.97 m	200.0 m	259.1 u	573.1 m	42.04 m
300.0 m	127.7 u	493.0 m	41.97 m	300.0 m	259.1 u	573.1 m	42.04 m
400.0 m	127.7 u	493.0 m	41.97 m	400.0 m	259.1 u	573.1 m	42.04 m
500.0 m	127.7 u	493.0 m	41.97 m	500.0 m	259.1 u	573.1 m	42.04 m
600.0 m	127.7 u	493.0 m	41.97 m	600.0 m	259.1 u	573.1 m	42.04 m
700.0 m	127.7 u	493.0 m	41.97 m	700.0 m	259.1 u	573.1 m	42.04 m
800.0 m	127.7 u	493.0 m	41.97 m	800.0 m	259.1 u	573.1 m	42.04 m
900.0 m	127.7 u	493.0 m	41.97 m	900.0 m	259.1 u	573.1 m	42.04 m
1000.0 m	127.7 u	493.0 m	41.97 m	1000.0 m	259.1 u	573.1 m	42.04 m
1.100	127.7 u	493.0 m	41.97 m	1.100	259.1 u	573.1 m	42.04 m
1.200	127.7 u	493.0 m	41.97 m	1.200	259.1 u	573.1 m	42.04 m
1.300	127.7 u	493.0 m	41.97 m	1.300	259.1 u	573.1 m	42.04 m
1.400	127.7 u	493.0 m	41.97 m	1.400	259.1 u	573.1 m	42.04 m
1.500	127.7 u	493.0 m	41.97 m	1.500	259.1 u	573.1 m	42.04 m
1.600	127.7 u	493.0 m	41.97 m	1.600	259.1 u	573.1 m	42.04 m
1.700	127.7 u	493.0 m	41.97 m	1.700	259.1 u	573.1 m	42.04 m
1.800	127.7 u	493.0 m	41.97 m	1.800	259.1 u	573.1 m	42.04 m

Figure 6: The tables to check whether the devices are in saturation region for different input DC levels

C. Plot the amplitude and phase of the frequency response.Analyze and explain whether your circuit is stable based on the plotted results.

I set the input DC voltage to 0^V and then plotted the Bode diagram of the circuit frequency response. I is represented in figure 7.

As depicted in the figure above, at the frequency of Px(1.164^{GHz}) the circuit still has a positive gain of $10.45dB$. Therefore, the uncompensated circuit is not stable under unity negative feedback.

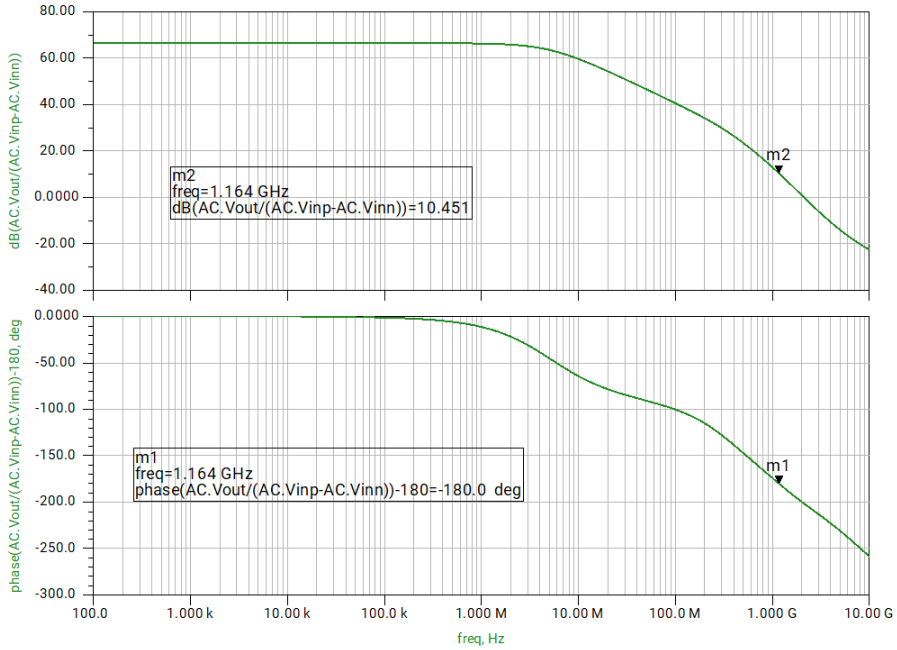


Figure 7: The Bode diagram of the circuit frequency response

d. Calculate the theoretical gain using g_m and r_o obtained from *ADS* operating points, and compare it with the simulation result.

First we derive the small-signal parameters in simulation.

index	Gm	1/Gds
1	142.3 u	287.4 k
2	907.7 u	47.57 k
3	920.9 u	57.63 k
4	1.951 m	26.24 k
5	1.951 m	26.24 k
6	2.040 m	42.72 k
7	2.040 m	42.72 k
8	4.128 m	22.69 k

Figure 8: The small-signal parameters of the transistors

The r_o of the devices is measured by $\frac{1}{G_{ds}}$ in *ADS*.

Using the derived values and the gain derived in part a., we calculate the gain of the circuit in low frequencies.

$$A_v = 1.951^{m\mho}(26.24^{k\Omega}||42.72^{k\Omega}) \times 4.128^{m\mho}(57.63^{k\Omega}||22.69^{k\Omega}) \approx 2131.35 = 66.57dB$$

The calculated theoretical gain is approximately the same as the one we got using simulation.

e. Find the feedback coefficients theoretically that make the system stable.

According to the Bode diagram of figure 7, the maximum feedback for stability is:

$$20\log(\frac{1}{\beta_{max}}) = 10.451 \Rightarrow \beta_{max} \approx 0.3$$

I plotted $\beta H(s)$ for beta = [0.1, 0.2, 0.3, 0.5] and observed the result.

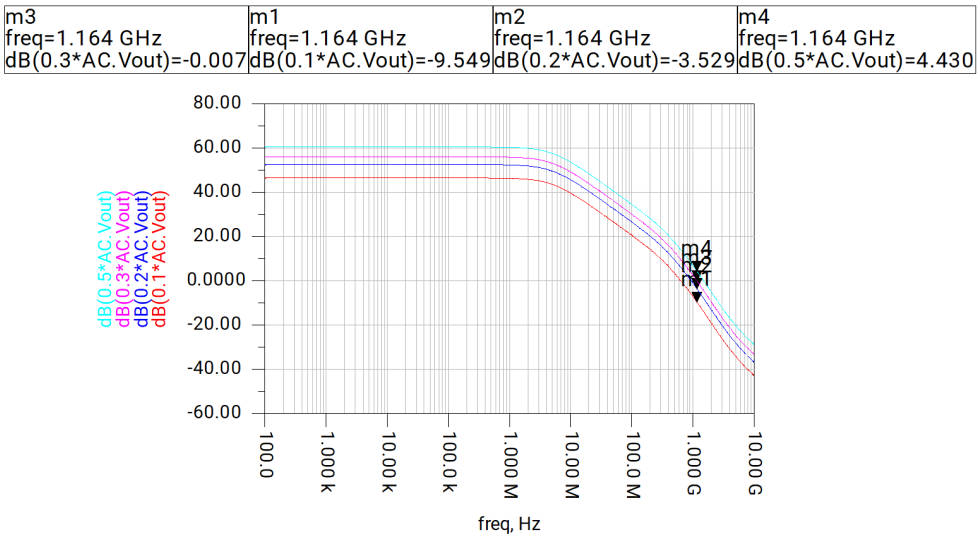


Figure 9: The frequency response for feedback = [0.1, 0.2, 0.3, 0.5]

We can see that for $\beta = 0.3$ the gain in frequency of Px is less than $0dB$. So we can say that for $\beta = 0.3$ the circuit will be in the edge of stability and the theoretical calculations are correct. For any value of feedback less than this, the circuit would be stable.

(I actually tried to sweep the feedback coefficient β, but the quality of the results I got wasn't so good and it wasn't clear where the gain actually reaches $0dB$. Eventually after searching for other approaches, I found out that I can use Tuner to find the right amount of feedback but it wasn't necessary in this part and I used Tuner in the following parts.)

f. Compensate your circuit using a single load capacitor to achieve a phase margin of 60° with a unity feedback coefficient, and report the required capacitor value. Then, plot the amplitude and phase of the frequency response, and report the phase margin.

The output resistance of the first and second stage is nearly the same so we can't tell which node the dominant pole is associated with. So I first put a capacitor in the output node and then a capacitor in the output node of the first stage and compared the results.

In order to find the right amount of capacitor in each node I used a variable C_c and Tuner so that I could tune the value of the capacitor and see the effect on the result.

For the output node the value of the capacitor needed to reach a Phase Margin of 60°was $C_c = 1.163^{nF}$ which is a really large capacitor.

The Tuner setting and the Bode diagram are shown in figures 10 and 12 respectively.

For the output node of the first stage on the other hand, only a $C_c = 347.6^{pF}$ capacitor is needed which is much better than the latter one.

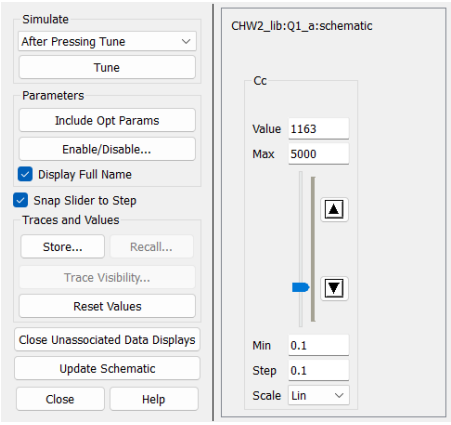


Figure 10: The setting of the Tuner

The Tuner setting and the Bode diagram are shown in figures 11 and 13 respectively.

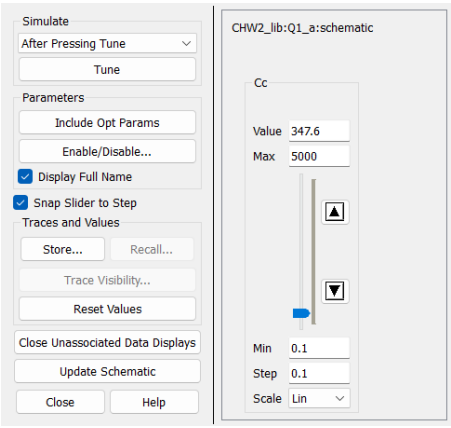


Figure 11: The setting of the Tuner

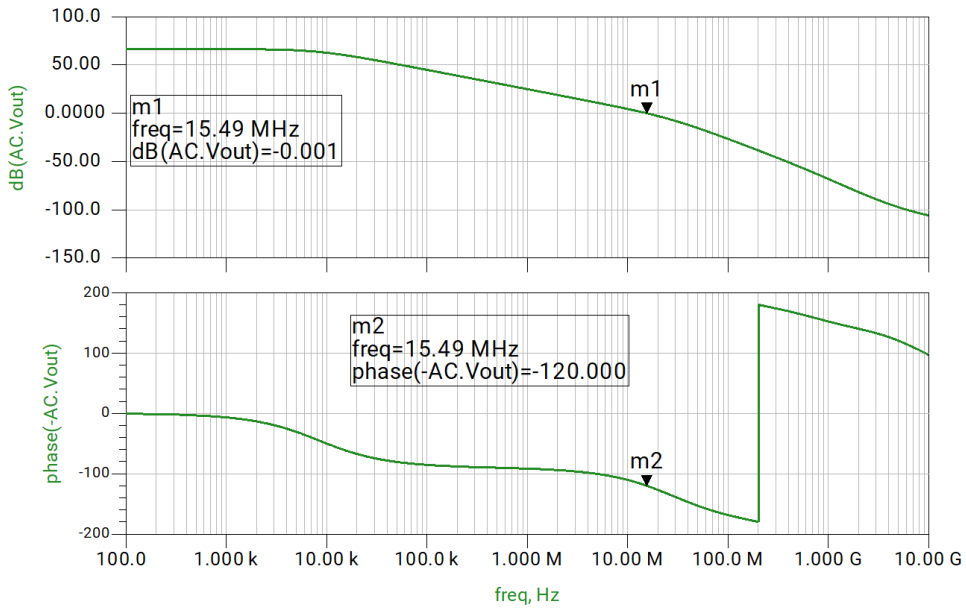


Figure 12: The Bode diagram of the circuit compensated by output node capacitor $C_c = 1.163^{nF}$

In conclusion, I preferred the second approach and added the capacitor to the output node of the first stage because it was smaller than the other one and the smaller the capacitor

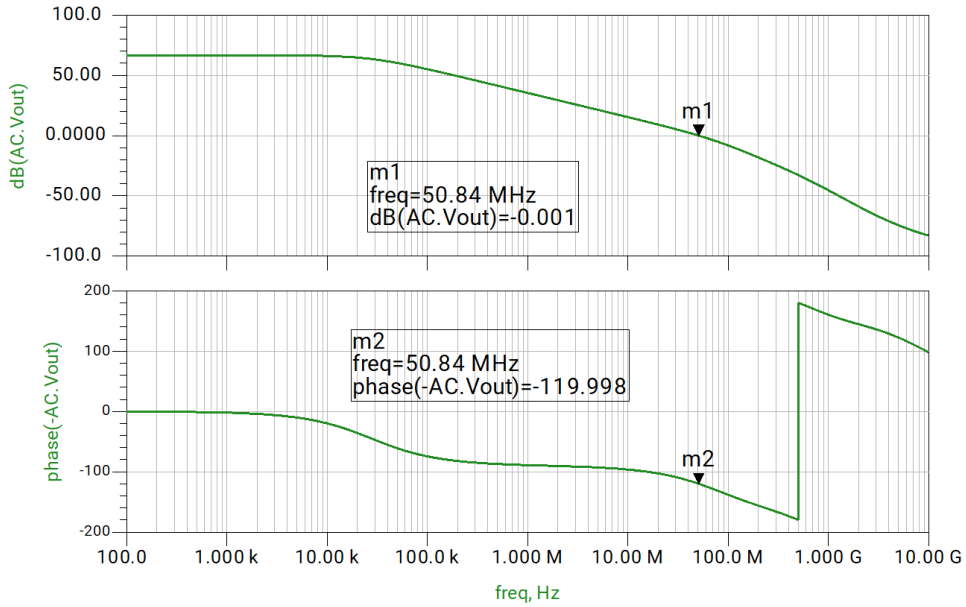


Figure 13: The Bode diagram of the circuit compensated by output node capacitor $C_c = 347.6^{pF}$

the smaller and better the IC.

Another thing to notice is that the Unity-gain bandwidth in the latter procedure is nearly 3 times the one in the first approach.

The designed circuit is demonstrated in the figure below.

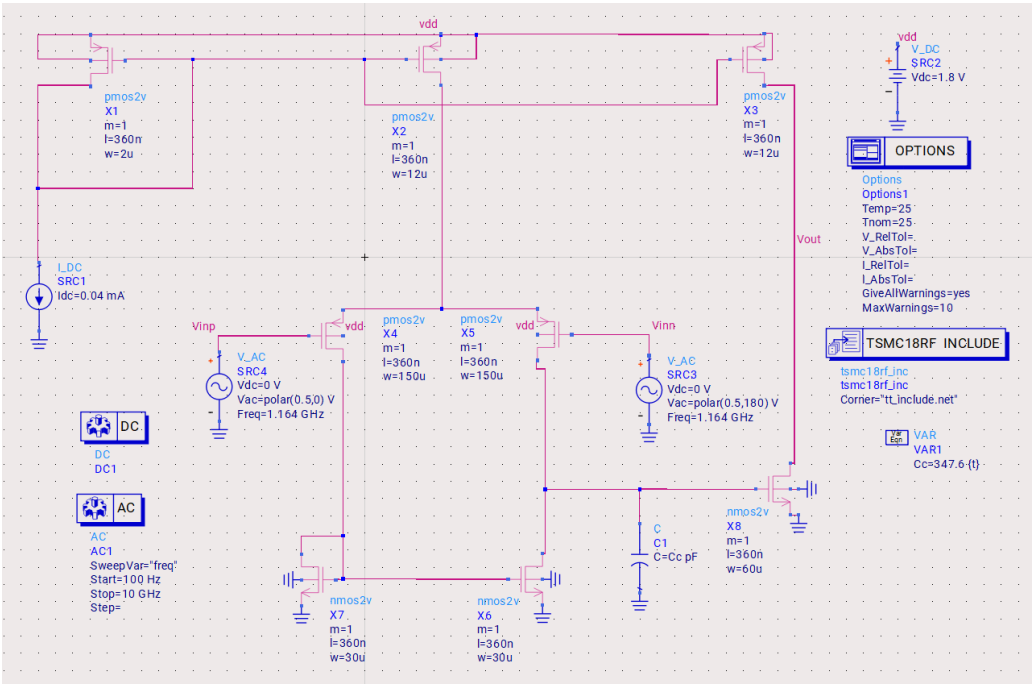


Figure 14: The designed circuit for part f

g. Insert a compensation capacitor between the first and second stages.(You may also use a series resistor if needed.) Determine its value to satisfy the following specifications. Then, plot both the amplitude and phase of the frequency response, and report the resulting phase margin and gain margin:

- Phase margin $> 60°$
- Unity-gain bandwidth $f_u > 200\,\text{MHz}$

In this part we need to insert a capacitor between the output node of the circuit and the output node of the first stage to use the Miller's effect at both nodes.

To do this I used a capacitor in series with a resistor between the input and output nodes of the second stage. To meet the specifications described, the first step is to find where each zero and pole of the circuit is.

To find the poles, as taught in the tutorial video, I drew lines with slope $0dB/dec$, $-20dB/dec$, $-40dB/dec$, ... tangent to the frequency response of the circuit. In this way we approximate the frequency response with some straight lines. The poles and zeros of the approximated frequency response would be the intersection of the consecutive lines.

In the following figure you can see the procedure.

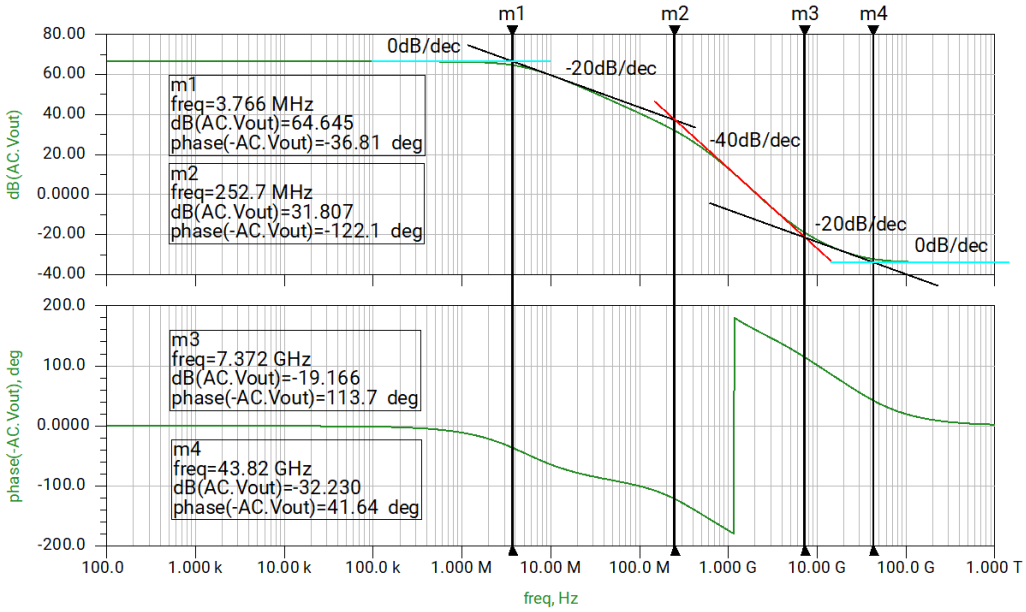


Figure 15

From the approach used above I found the poles and zeros as below:

$$P_1 = P_{dom} = 3.766^{MHz}, P_2 = 252.7^{MHz}, Z_1 = 7.372^{MHz}, Z_2 = 43.82^{MHz}$$

When we insert the capacitor because of the Miller's effect, the dominant pole will become extremely small that we can assume that in the frequencies near the second pole the phase is already dropped by 90°.

To have a Phase Margin of 60°there should be a 30°decrease in phase from $0.1P_2$ to G_x.

The specifications state that G_x should be larger than 200^{MHz}. So I assumed $G_x = 250^{MHz}$ to have a safety margin from the limit.

$$G_x = 250^{MHz} \rightarrow 45\log(\frac{G_x}{0.1P_{2_{new}}}) = 30 \Rightarrow P_{2_{new}} = 538.61^{MHz}$$
$$\Rightarrow 66.52 - 20\log(\frac{G_x}{P_{1_{new}}}) = 0 \Rightarrow P_{1_{new}} \approx 118.02^{kHz}$$

During the course we learned that when we put a compensating capacitor in series with a resistor between two stages, the new values of the poles would be:

$$P_{1_{new}} = \frac{1}{2\pi R_1(g_{m_2}R_2)C_c}, P_{2_{new}} = \frac{g_{m_2}C_c}{2\pi(C_1C_2 + C_1C_c + C_2C_c)}$$
$$R_1 = 26.24^{k\Omega}||42.72^{k\Omega} = 16.255^{k\Omega}, R_2 = 57.63^{k\Omega}||22.69^{k\Omega} = 16.28^{k\Omega}$$
$$P_{1_{old}} \approx \frac{1}{2\pi R_1C_1} \Rightarrow C_1 \approx 2.604^{pF}, P_{2_{old}} = \frac{1}{2\pi R_2C_2} \Rightarrow C_2 \approx 0.039^{pF}$$
$$C_c = \frac{1}{2\pi R_1 g_{m_2} R_2 P_{1_{new}}} \approx 1.235^{pF}$$

This approach also introduces a new pair of pole and zero to the frequency response which are calculated using:

$$P_3 = \frac{C_1C_2 + C_1C_c + C_2C_c}{2\pi R_zC_1C_2C_c}, Z = \frac{1}{2\pi(R_z - \frac{1}{g_{m_2}})C_c}$$

I choose R_z in a way that the zero would be ∞ so that it doesn't affect the circuit's response.

$$R_z = \frac{1}{g_{m_2}} \approx 242.25^{\Omega} \rightarrow P_3 \approx 17.63^{GHz}$$

Now that I have determined the values of the capacitor and resistor, I simulate the compensated circuit.

The results are as below.

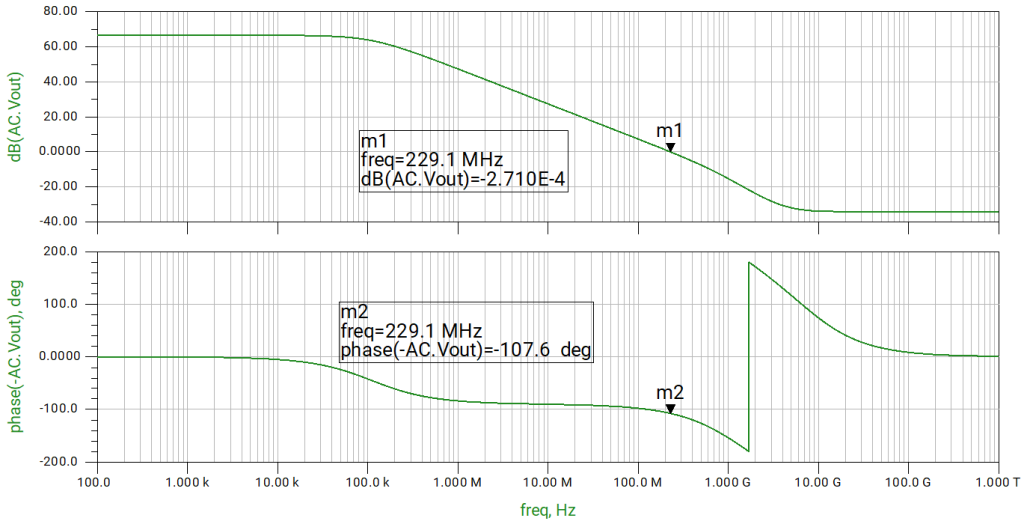


Figure 16: The Bode diagram of the compensated circuit with $C_c = 1.235^{pF}, R_z = 242.25^{\Omega}$

As demonstrated in the figure above, the Phase Margin is nearly 72°and also the Unity-gain bandwidth is $229.1MHz$ which is above the specified limit.

But I observed that I still can tune the value of the capacitor in a way that the Unity-gain bandwidth increases and we still have a Phase Margin greater than 60°.

I used the method in the previous part to tune the capacitor and eventually after trying several values for the capacitor I found that for $C_c = 0.8^{pF}$ we achieve a Unity-gain bandwidth of $338.9MHz$ while the Phase Margin is about 64°.

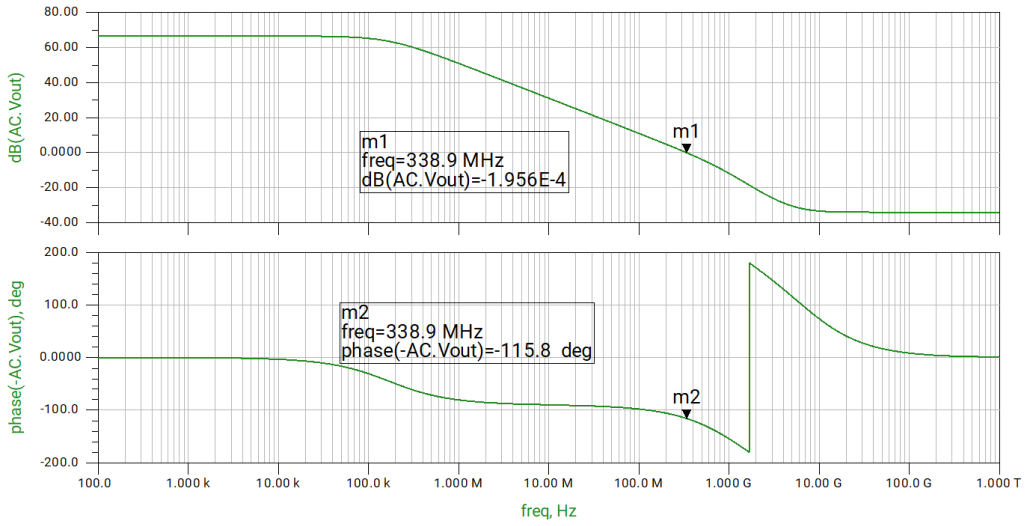


Figure 17: The Bode diagram of the compensated circuit with $C_c = 0.8^{pF}, R_z = 242.25^{\Omega}$

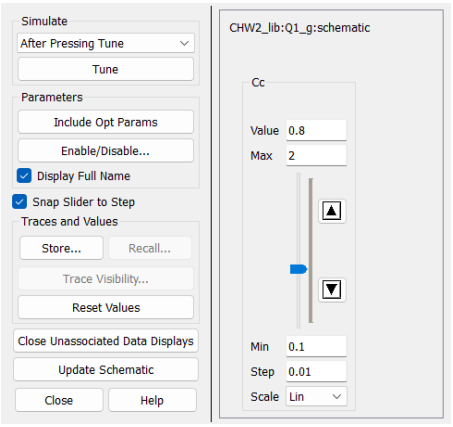


Figure 18: The setting of the Tuner

The final designed circuit is shown in figure 19.

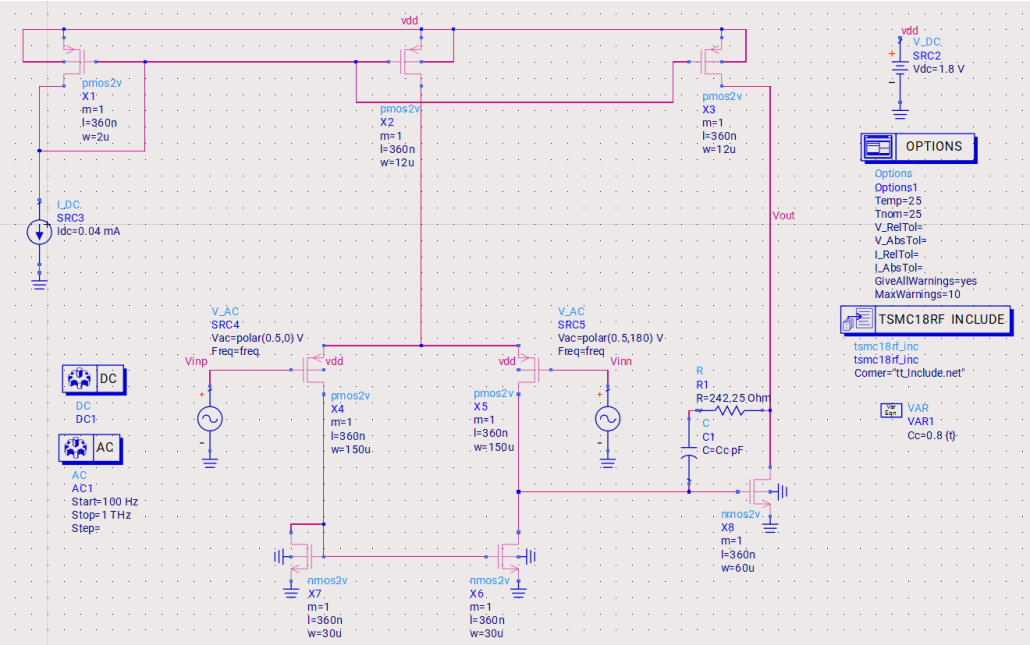


Figure 19: The final designed circuit with compensation

h. Using a common-gate stage for compensation with a phase margin of 60°, design the required circuit. Then, plot the amplitude and phase of the output and report the phase margin.(bonus)

i. Compute the slew rate of the circuit using *ADS*.

In order to measure the slew rate of our circuit, We should give one of the inputs a rectangular pulse with a large amplitude(1.8^V here) and simulate the circuits response using Transient Analysis.

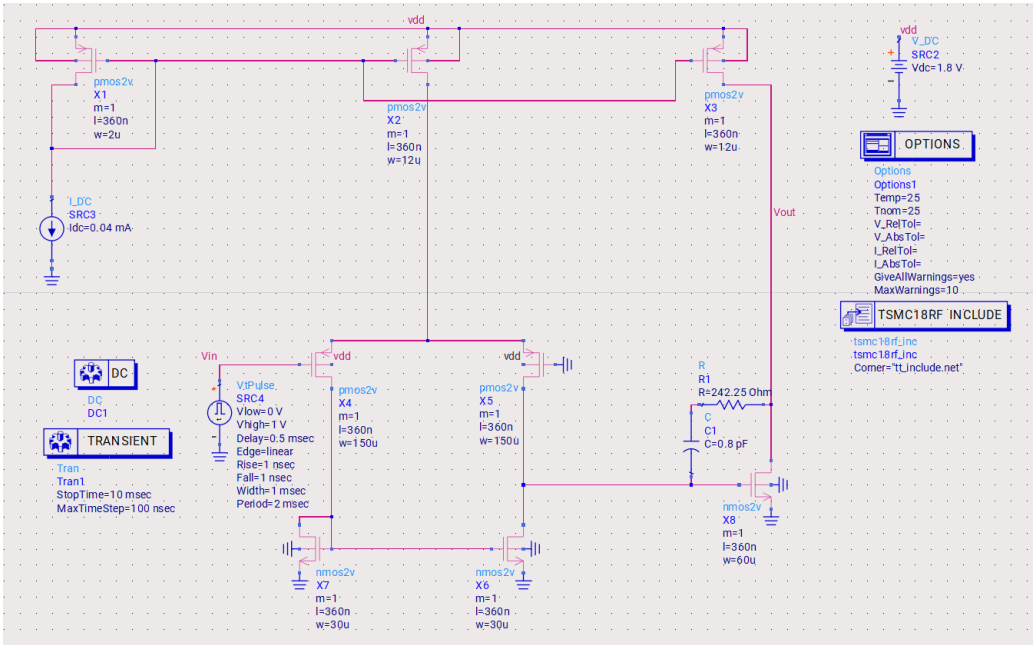


Figure 20: The configurations needed to measure slew rate

The circuit needed in this part is displayed in figure 20.

To measure the slew rate I first plotted the output signal in time domain and then zoomed in to measure the slope of the signal going from the minimum voltage to the maximum voltage.

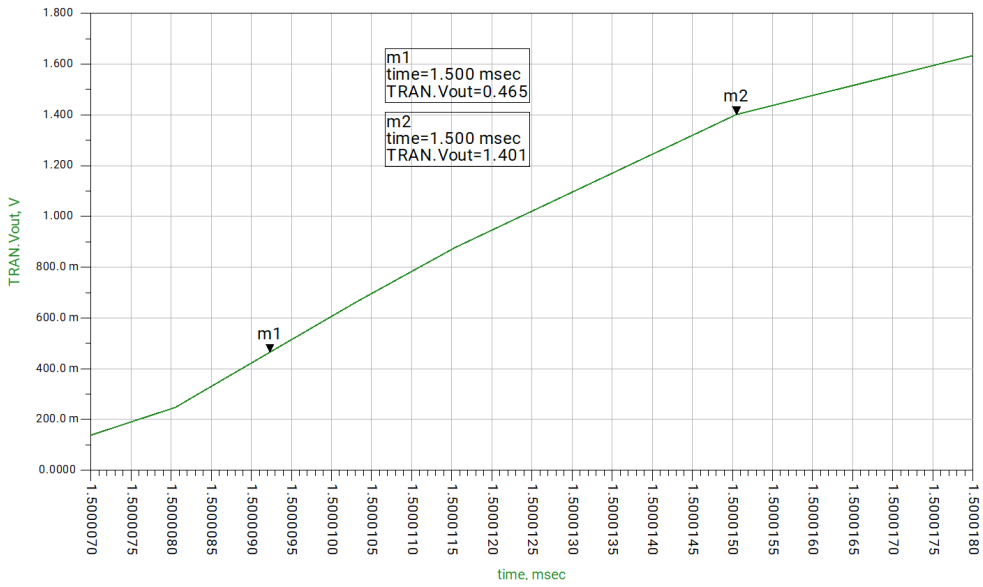


Figure 21: The transition of the output voltage

As depicted in the preceding figure the slew rate of the circuit is:

$$\text{slew rate} = \frac{1.401^V - 0.465^V}{1.500015ms - 1.50000925ms} \approx 162.8V/\mu s$$

j. Plot the time-domain output using transient analysis in *ADS* and report the output symmetric swing.

In the last part of this assignment we would like to analyze the swing of our designed circuit. To achieve the gain specification I designed the circuit i a way that V_{od} of M_3 became something about $0.4^V - 0.5^V$. So the maximum output voltage level possible we can get from this circuit is $1.8^V - 0.5^V = 1.3^V$. Ignoring the effect of the compensation capacitor on the swing, the minimum output voltage level is nearly 0.1^V.

Knowing these I simulated the circuit giving sine voltages to both inputs with opposite polarities. The gain of the circuit was approximately 2130 so considering 1.2^V swing, the starting amplitude of the input signals would be 0.55^{mV}

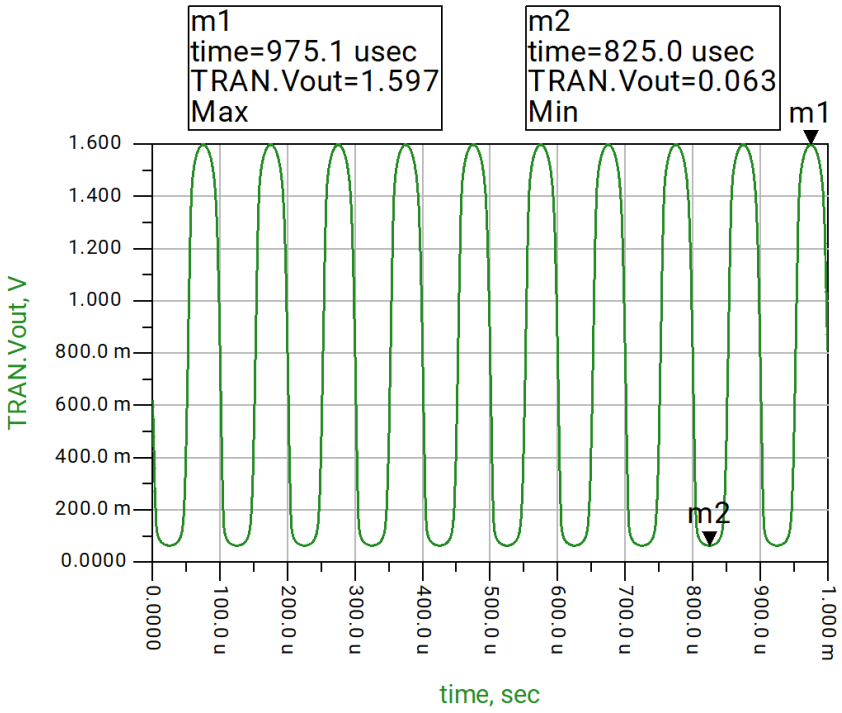


Figure 22: The output signal corresponding to the input signal amplitude 0.55^{mV}

As it can be observed above, the circuit does not work properly and we need to lower the amplitude of the input signal. Testing a few amplitudes, the final value for the input signal amplitude was $125^{\mu V}$.

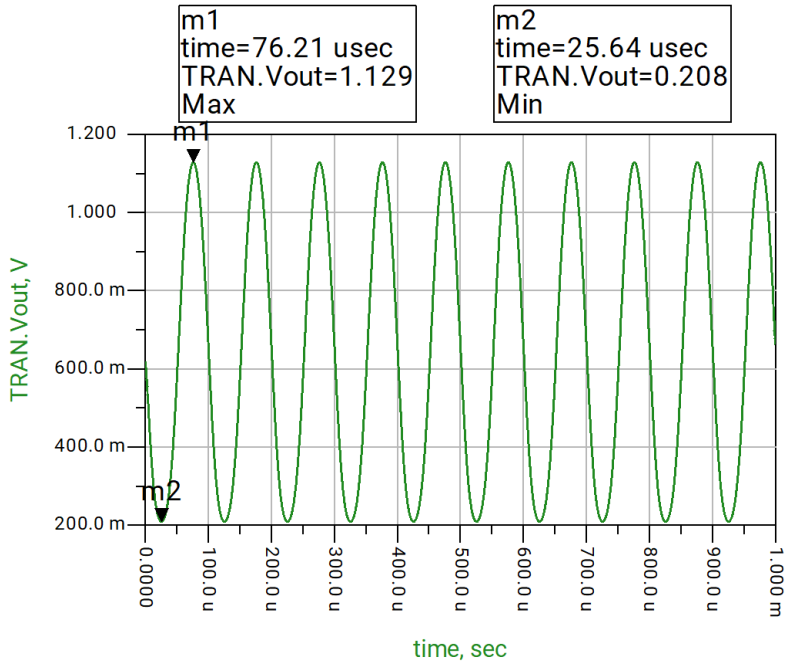


Figure 23: The maximum output swing of the circuit

The maximum swing of the circuit was measured $1.129^{V} - 0.208^{V} = 0.92^{V}$ which is a good swing for the circuit with the specifications mentioned above.